

Defense sheet — Assignment set 1

Scope: this sheet covers the **coded** sub-questions only (2b, 2d, 2e, 3a, 3b, 3c). The hand-derivation parts (1a, 1b, 1c, 2a, 2c) are in the PDF and are not repeated here, but Fang can jump from code to derivation, so the cross-references below matter.

The task (in my own words)

Assignment 1 has three blocks. The graded coded pieces sit in blocks 2 and 3.

- **Block 2 — the COS method, sine variant.** The paper builds COS from a Fourier-**cosine** expansion; the assignment asks me to redo it with a Fourier-**sine** expansion.
 - **2b:** program the sine version of eq. (11) and reproduce the density-recovery errors of Table 1 of the COS paper for the standard normal density.
 - **2d:** implement the sine-COS European **put** pricing formula (derived by hand in 2c) under Black-Scholes parameters $S_0=3$, $\sigma=0.3$, $T=1$, $K=4$, $r=0.03$, and report the absolute error vs the analytic BS put as a function of N .
 - **2e:** extend the sine recovery one integration step further to recover the **CDF**, then verify it against the analytic log-normal CDF.
- **Block 3 — importance sampling for a tail quantile.** Portfolio of $N=1000$ digital options; $L_T = \sum_i 1\{S_i(T) > K_i\}$ with a common Brownian factor W and idiosyncratic factors B_i . Estimate the 95% quantile of L_T .
 - **3a:** plain Monte Carlo.
 - **3b:** importance sampling, mean shift $W_T \sim N(1.5, T)$, reweight by p/q .
 - **3c:** importance sampling, mean + variance shift $W_T \sim N(1.5, 2T)$, reweight by p/q .

TODO (Bruno): In one sentence each, state the *point* of each sub-question — e.g. why 2b uses the standard normal density specifically, why 2e is “one extra integration step” beyond 2b/2c, and what the 3b-vs-3c comparison is meant to demonstrate about importance sampling.

My approach and why

What the code factually does (per block):

- **2b / 2e** build the sine series coefficients $F_k = (2/(b-a)) * \text{Im}\{\phi(w_k) \exp(-i w_k a)\}$ with $w_k = k\pi/(b-a)$, then sum F_k against either $\sin(k\pi(x-a)/(b-a))$ (density, 2b) or the integrated sine $(1 - \cos(\dots))$ form (CDF, 2e).
- **2d** prices the put as $e^{-rT} * \sum_k \text{Im}\{\phi(w_k) \exp(-i w_k a)\} * V_k$, where V_k is the analytic sine coefficient of the put payoff built from the ϕ_k/χ_k primitives.
- **3a/3b/3c** simulate $S_i(T)$ from the exact log-normal solution, count threshold crossings to form L_T , and take a (weighted, in 3b/3c) empirical 95% quantile.

TODO (Bruno): Justify, in your own words, the *modelling* choices that the code hard-codes: - 2b: why $[a,b] = [-10, 10]$ is a safe truncation for the standard normal. - 2d: why you used the **cumulant-based** range $[a,b] = c1 \pm L*\sqrt{c2 + \sqrt{c4}}$ with $L=10$, and why that is the “correct” way to set the range rather than a fixed window. - 2e: why you switched to a **fixed** window $[a,b] = [-2, 2]$ for the CDF test instead of reusing the cumulant rule from 2d. - 3b/3c: why a mean shift of exactly 1.5, and why doubling the variance is the natural “second” scheme to test.

Key code sections (file + what it does + why I wrote it that way)

2b — 2bCF (2).py (sine density recovery, standard normal)

- 2bCF (2).py:4-5 set the truncation range $a=-10$, $b=10$.

- 2bCF (2).py:6-7 set the evaluation grid $X_{\text{test}} = \text{arange}(-5,6)$ (i.e. integers -5..5) and $N_{\text{vals}} = [4,8,16,32,64]$.
- 2bCF (2).py:10-12 $\text{normal_density}(x)$ returns the standard normal pdf $(1/\sqrt{2\pi}) \exp(-x^2/2)$.
- 2bCF (2).py:14-16 $\text{char_function}(w)$ returns $\exp(-0.5 w^2)$, the characteristic function of $N(0,1)$.
- 2bCF (2).py:18-33 $\text{sin_density}(x_{\text{vals}}, a, b, N)$:
 - :22 $k = \text{arange}(1, N+1)$ — sine series starts at $k=1$ (no $k=0$ term).
 - :23 $w_k = k\pi/(b-a)$.
 - :26 $f_k = (2/(b-a)) * \text{Im}\{ \text{char}(w_k) * \exp(-i w_k a) \}$ — the sine coefficients F_k .
 - :30-31 for each x , forms $\sin(k \pi (x-a)/(b-a))$ and sums $f_k * \sin_{\text{terms}}$.
- 2bCF (2).py:39-43 loops over N , computes $\max|f_{\text{approx}} - f_{\text{true}}|$ and prints it.

TODO (Bruno): Explain *why* line 26 takes the **imaginary** part (tie it to the sine series and to Euler's formula in your 2a derivation), and why the coefficient prefactor is $2/(b-a)$ and not $1/(b-a)$.

2d — 2dCF (2).py (sine-COS European put pricing)

FACT / FORMATTING FLAG: this file is stored as a **single physical line** (the newlines were stripped). So I cannot give honest per-line numbers here — I reference it by **function name**. Fang may notice this; be ready to say it is a save/encoding artefact, not the logic.

- Parameters block: $S_0=3$, $\sigma=0.3$, $T=1$, $K=4$, $r=0.03$; $x_0 = \log(S_0/K)$; $\mu = r - 0.5 \sigma^2$.
- Cumulant range: $c_1 = x_0 + \mu T$, $c_2 = T\sigma^2$, $c_4 = 0$, $L = 10$, $a = c_1 - L\sqrt{c_2 + \sqrt{c_4}}$, $b = c_1 + L\sqrt{c_2 + \sqrt{c_4}}$ (the comment cites “eq 49”).
- $\text{phi_BS}(w, x, r, \sigma, T)$: characteristic function of $y = \ln(S_T/K)$, returns $\exp(i w \mu_y - 0.5 \text{var}_y w^2)$ with $\mu_y = x + (r - 0.5 \sigma^2) T$, $\text{var}_y = \sigma^2 T$.
- $\text{phi}_k(k, a, b, c, d)$: returns $(1/w_k) * (\cos(w_k(c-a)) - \cos(w_k(d-a)))$ — the psi_k primitive from 2c.
- $\text{chi}_k(k, a, b, c, d)$: returns $\text{term}(d) - \text{term}(c)$ with $\text{term}(y) = e^y (\sin(w_k(y-a)) - w_k \cos(w_k(y-a))) / (1 + w_k^2)$ — the chi_k primitive from 2c.
- $\text{V}_k\text{put}(a, b, K, k_s)$: returns $(2K/(b-a)) * (-\text{chi}_k(k_s, a, b, a, 0) + \text{phi}_k(k_s, a, b, a, 0))$, i.e. the payoff coefficient $V_k = (2K/(b-a))(\text{psi}_k(a, 0) - \text{chi}_k(a, 0))$.
- $\text{sin_put_price}(N, \dots)$: builds $\text{cf_part} = \text{Im}\{ \text{phi_BS}(w_k, x_0, \dots) \exp(-i w_k a) \}$, multiplies by V_k , and returns $\exp(-rT) * \text{dot}(\text{cf_part}, V_k)$.
- $\text{bs_put}(\dots)$: closed-form BS put for the reference price.
- The driver loops N in $[4,8,16,32,64,128,256,512,1024]$ and prints $|\text{SIN} - \text{BS}|$.

NAMING FLAG (be ready for this): in the report (2c) psi_k is the integral of $\sin$ and chi_k the integral of $e^y \sin$. In this file the **function named phi_k** is the $\sin$ integral (the report's psi_k) and the function named chi_k is the $e^y \sin$ integral. So phi_k here = psi_k in the report. The numbers still match because V_kput pairs them correctly, but the name phi_k collides with the characteristic function phi .

TODO (Bruno): Justify the *content* choices: why $c_4 = 0$ is acceptable here (what is the 4th cumulant of a Gaussian log-price?), why $x_0 = \log(S_0/K)$ is the right state variable, and why the discount factor is e^{-rT} rather than $e^{-r\Delta t}$ with some other Δt .

2e — 2eCF (2).py (sine-COS CDF recovery)

- 2eCF (2).py:5-8 parameters $r=0.03$, $S_0=3$, $\sigma=0.3$, $T=1$.
- 2eCF (2).py:11-12 $\mu = (r - 0.5 \sigma^2) T$, $\sigma_{\text{aux}} = \sigma\sqrt{T}$ — moments of $\ln(S_T/S_0)$.
- 2eCF (2).py:14-15 fixed range $a=-2$, $b=2$.
- 2eCF (2).py:16-17 $X_{\text{test}} = \text{linspace}(-0.5, 0.5, 11)$, $N_{\text{vals}} = [4,8,16,32,64,128]$.
- 2eCF (2).py:19-23 $\text{char_function}(w) = \text{CF of } \ln(S_T/S_0)$: $\exp(i w \mu - 0.5 \sigma_{\text{aux}}^2 w^2)$.

- 2eCF (2).py:25-29 $F_{\text{exact}}(z) = \text{norm.cdf}((z - \mu)/\sigma_{\text{aux}})$ — analytic normal CDF.
- 2eCF (2).py:31-46 $\text{sin_cdf}(z_vals, N, a, b)$:
 - :36-37 $ks = \text{arange}(1, N+1)$, $wk = ks \pi / (b-a)$.
 - :40 $im = \text{Im}\{ \exp(-i wk a) \}$.
 - :43 $\text{cosine} = \cos(\text{outer}((z-a)/(b-a), ks) * \pi)$ — matrix of $\cos(k \pi (z-a)/(b-a))$.
 - :46 returns $(2/\pi) * (im/ks) @ (1 - \text{cosine}).T$, i.e. $F(z) \approx (2/\pi) \sum_k (1/k) \text{Im}_k (1 - \cos(k \pi (z-a)/(b-a)))$.
- 2eCF (2).py:54-57 loop over N , print $\max|F_{\text{sin}} - F_{\text{true}}|$.

TODO (Bruno): Explain *why* the CDF formula carries a $1/k$ factor and a $2/\pi$ prefactor (where do they come from when you integrate the density term by term?), and why the $(1 - \cos(\dots))$ shape is the integrated form of $\sin(\dots)$.

3a — 3aCF (2).py (plain Monte Carlo quantile)

- 3aCF (2).py:5-12 parameters $N_{\text{stocks}}=1000$, $S_0=1$, $r=0.01$, $\sigma_1=0.8$, $\sigma_2=0.6$, $T=1$, $K=2$, $\alpha = 1-0.95$.
- 3aCF (2).py:14 $\text{drift} = r - 0.5 * (\sigma_1^2 + \sigma_2^2)$.
- 3aCF (2).py:16-31 $\text{simulate_plainMC}(\text{paths})$:
 - :22 common factor $W_t \sim N(0, \sqrt{T})$ (shape paths).
 - :23 idiosyncratic $B_i \sim N(0, \sqrt{T})$ (shape paths $\times$ N_{stocks}).
 - :26 $S_T = S_0 \exp(\text{drift} * T + \sigma_1 * W_t[:, \text{None}] + \sigma_2 * B_i)$.
 - :29 $L_T = \text{sum}(S_T > K, \text{axis}=1)$ — count of in-the-money names per path.
- 3aCF (2).py:36 $\text{np.random.seed}(42)$.
- 3aCF (2).py:40-44 loops paths over $[1000..500000]$, $q = \text{np.quantile}(L_T, 1-\alpha) = 0.95$ quantile.
- 3aCF (2).py:47-55 semilog convergence plot.

TODO (Bruno): Justify why $\text{np.std}/\text{np.random.normal}(0, \sqrt{T})$ (a $N(0, T)$ draw) is the *exact* terminal law of W_T here, why no time-stepping is needed, and why the empirical quantile is taken at $1 - \alpha = 0.95$ given the assignment's definition $P(L_T \geq q_\alpha) \leq \alpha$.

3b — 3bCF (2).py (importance sampling, mean shift)

- 3bCF (2).py:17-18 IS parameters $\mu_{\text{shift}} = 1.5$, $\mu_{\text{shiftsq}} = \mu_{\text{shift}}^2 / (2T)$.
- 3bCF (2).py:20-34 $\text{weighted_quantile}(\text{values}, \text{weights}, q)$:
 - :25-27 sort values, reorder weights to match.
 - :30 $\text{cdf} = \text{cumsum}(w_{\text{sorted}}) / \text{sum}(w_{\text{sorted}})$.
 - :33-34 $\text{idx} = \text{searchsorted}(\text{cdf}, q)$, return $v_{\text{sorted}}[\text{idx}]$.
- 3bCF (2).py:43 $W_q \sim N(\mu_{\text{shift}}, \sqrt{T})$ — common factor drawn under q .
- 3bCF (2).py:44 $B_i \sim N(0, \sqrt{T})$ — idiosyncratic factors unchanged.
- 3bCF (2).py:47 $S_T = \exp(\text{drift} * T + \sigma_1 * W_q[:, \text{None}] + \sigma_2 * B_i)$ (note: no explicit S_0 , since $S_0=1$).
- 3bCF (2).py:50 $L_T = \text{sum}(S_T > K, \text{axis}=1)$.
- 3bCF (2).py:53 $\text{ratio} = \exp(-(\mu_{\text{shift}}/T) * W_q + \mu_{\text{shiftsq}})$ — likelihood ratio p/q .
- 3bCF (2).py:56 $q_{.95} = \text{weighted_quantile}(L_T, \text{ratio}, 0.95)$.

TODO (Bruno): Derive the ratio on line 53 from p/q of two Gaussians and confirm it equals the report's $\exp(-1.5 w + 1.125)$ at $T=1$. Also justify *why only W is reweighted and not B_i* (the report says the B_i ratio is 1 — state precisely why).

3c — 3cCF (2).py (importance sampling, mean + variance shift)

- 3cCF (2).py:17-18 IS parameters $\mu_{\text{shift}} = 1.5$, $\text{var}_q = 2 * T$.
- 3cCF (2).py:20-34 weighted_quantile — identical structure to 3b.
- 3cCF (2).py:43 $W_q \sim N(\mu_{\text{shift}}, \sqrt{\text{var}_q}) = N(1.5, 2T)$.
- 3cCF (2).py:44, 47, 50 same B_i , S_T , L_T construction as 3b.

- `3cCF (2).py:54 ratio = sqrt(2)*exp((-W_q^2 - 3 W_q + 2.25)/(4T))` — the p/q for the mean+var shift.
- `3cCF (2).py:56 q_95 = weighted_quantile(L_T, ratio, 0.95).`

TODO (Bruno): Derive the line-54 ratio from $p \sim N(0, T)$, $q \sim N(1.5, 2T)$ and confirm the `sqrt(2)` prefactor comes from the variance mismatch in the normalising constants. Then explain why this scheme is *worse* for the 95% quantile than 3b (the report’s argument about mass landing too far in the tail) — restate it as your own.

Design decisions I must justify

- **Sine vs cosine expansion.** The code implements the sine series throughout (coefficients via `Im{...}`). **TODO (Bruno):** why does the assignment want sine, and what changes structurally vs the cosine COS (the $k=0$ half-weight term, the Im vs Re)?
 - **Truncation range, three different choices.** 2b uses `[-10,10]`, 2d uses the cumulant rule with `L=10`, 2e uses `[-2,2]`. **TODO (Bruno):** justify each range and explain whether `[-2,2]` in 2e is wide enough for $N(\mu, \sigma_{\text{aux}}^2)$ with $\mu = -0.015$, $\sigma_{\text{aux}} = 0.3$ (it is $\sim \pm 6.7$ sigma — confirm and say so). This is exactly the kind of “is your range safe?” question Fang asks.
 - **$L = 10$ cumulant multiplier.** **TODO (Bruno):** the paper uses L around 8-10; justify your $L=10$ and what error you’d see if L were too small (link to your Assignment 2 short-maturity / small- $c2$ bug).
 - **np.quantile interpolation vs the weighted_quantile searchsorted rule.** 3a uses NumPy’s `np.quantile` (linear interpolation); 3b/3c use a custom step-function `searchsorted` quantile. **TODO (Bruno):** why two different quantile estimators, and does the mismatch bias the 3a-vs-3b/3c comparison?
 - **seed(42).** Used in 3a/3b/3c for reproducibility. **TODO (Bruno):** confirm this is only for reproducibility and does not, e.g., reuse the same W across schemes in a way that matters.
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Results and what they mean

Reported in the PDF (factual, from the tables):

- **2b:** max density error falls $2.1e-1$ ($N=4$) $\rightarrow 1.0e-16$ ($N=64$) — matches the paper’s Table 1 order of magnitude; spectral convergence.
- **2d:** `|SIN - BS|` falls to $\sim 3.3e-16$ by $N=64$ and stays there; BS reference put = 0.9918191782.
- **2e:** max CDF error falls $1.37e-1$ ($N=4$) $\rightarrow 3.67e-11$ ($N=32$) and then **plateaus** at $3.67e-11$.
- **3a:** 95% quantile estimate settles near 590 as paths grows to $5e5$.
- **3b:** estimate converges to ~ 587 and visibly stabilises with far fewer paths than 3a.
- **3c:** estimate ends near 589 but the path is noisier; the report argues it is *less* efficient than 3b for this quantile.

TODO (Bruno): - 2d/2e: explain the **plateau** (why does the error stop improving — round-off vs truncation-range error?). For 2e specifically, why does it plateau at $3.67e-11$ and not at machine $1e-16$ like 2d? This is the single most likely “your convergence stalled — why?” question. - 3a/3b/3c: are 590 / 587 / 589 *consistent* (do their confidence intervals overlap)? What is the “true” quantile and how confident are you?

The hardest question Fang could ask here — and my answer

Likely hardest: “Your 2e CDF error stalls at $3.67e-11$ while your 2d price reaches $1e-16$. Both use the sine-COS machinery. Why the difference, and is $3.67e-11$ a bug or expected?”

This bites because it sits exactly on the truncation-range / convergence theme Fang co-authored, and the two tables disagree by five orders of magnitude.

TODO (Bruno): Write the answer. Components you must cover: 1. The CDF formula integrates the density term by term, so its error inherits the **truncation-range** error $f(a)$, $f(b) \neq 0$ at the endpoints — i.e. the residual comes from $[a,b]=[-2,2]$ not being infinitely wide, not from too few terms N . (Confirm by saying what happens if you widen $[a,b]$.) 2. Why 2d reaches machine precision instead: the put payoff coefficients V_k decay and the BS density is extremely well captured by the cumulant range, so series-truncation error drops below round-off. 3. State plainly whether $3.67e-11$ is acceptable or whether widening $[-2,2]$ would push it lower.

(Second candidate hardest: *“Derive line 53 / line 54 likelihood ratios from scratch and tell me why the variance shift in 3c hurts.”* — see the 3b/3c TODOs.)

“Show me in your code where you do X” — anticipated spots

- “...where you build the sine coefficients F_k .” -> 2bCF (2).py:26; same construction at 2eCF (2).py:40 and inside sin_put_price(cf_part) in 2dCF (2).py.
- “...where you set the COS truncation range.” -> 2bCF (2).py:4-5 (fixed), cumulant block in 2dCF (2).py (a = c1 - L*sqrt(c2+sqrt(c4))), 2eCF (2).py:14-15 (fixed $[-2,2]$).
- “...where the characteristic function enters pricing.” -> phi_BS and cf_part in 2dCF (2).py.
- “...where you compute the payoff coefficients V_k .” -> V_k_put (with phi_k/chi_k) in 2dCF (2).py.
- “...where you integrate the sine to get the CDF.” -> 2eCF (2).py:43-46 (the $(1 - \cosine)$ term).
- “...where you get the exact terminal stock price.” -> 3aCF (2).py:26, 3bCF (2).py:47, 3cCF (2).py:47.
- “...where you form L_T .” -> 3aCF (2).py:29, 3bCF (2).py:50, 3cCF (2).py:50.
- “...where you apply the likelihood ratio.” -> 3bCF (2).py:53, 3cCF (2).py:54.
- “...where you take the weighted quantile.” -> weighted_quantile in 3bCF (2).py:20-34 / 3cCF (2).py:20-34, called at :56.
- “...where you take the plain quantile.” -> 3aCF (2).py:42 (np.quantile(L_T , 1-alpha)).